DG INTERN HUB

SOC ANALYST INTERNSHIP

WEEK 2 Notes

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Topic: 10_Inbound_vs_Outbound_Traffic

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Inbound vs Outbound Traffic

Learning Objective

To understand the difference between inbound and outbound network traffic and learn how a SOC analyst can use this information during an investigation.

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1. What is Inbound Traffic?

Inbound traffic is network traffic that is coming **into** a device, server, or network from another system.

For example, if someone connects to a web server from the internet, the traffic coming to that server is inbound traffic.

Example

Internet

↓

Company Web Server

The traffic going from the internet to the company server is **inbound**.

Some examples include:

- Incoming SSH connection
- Incoming HTTP/HTTPS request
- Someone accessing a server
- Incoming connection attempts
- External system connecting to an internal host

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2. What is Outbound Traffic?

Outbound traffic is network traffic going **from** a device or network to another system.

For example, when I open a website from my laptop, my laptop sends traffic to the web server.

My Laptop

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Internet

That traffic leaving my laptop is **outbound**.

Some examples include:

- Browsing a website
- DNS queries
- Uploading files
- Sending emails
- A server connecting to another server
- A compromised machine communicating with an external IP

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3. Difference Between Inbound and Outbound Traffic

Inbound Traffic	Outbound Traffic
Traffic coming into a system/network	Traffic leaving a system/network
Source is usually outside the destination network	Source is usually inside the network
Example: Internet → Web Server	Example: Laptop → Internet
Can show incoming connection attempts	Can show connections made by internal systems

Useful for detecting unauthorized access attempts	Useful for detecting suspicious external communication

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4. Simple Example

Suppose a company has a server with IP:

192.168.1.10

If an external system connects to this server:

Internet → 192.168.1.10

This is **inbound traffic**.

If the same server connects to an external IP:

192.168.1.10 → Internet

This is **outbound traffic**.

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5. How Does a SOC Analyst Use This During an Investigation?

The direction of traffic can give useful context during an investigation.

Inbound investigation

If I see unusual inbound traffic, I might check:

- Source IP address
- Destination system
- Destination port
- Number of connection attempts
- Time of activity

- Whether the source IP is known
- Whether the connection was successful

For example, if an external IP is repeatedly trying to connect to SSH on a server, I would investigate whether it could be a brute-force attempt.

Outbound investigation

For suspicious outbound traffic, I might check:

- Destination IP
- Destination domain
- Destination port
- Process responsible for the connection
- User account
- Amount of data transferred
- Whether the destination is known or suspicious

For example, if a workstation that normally does not communicate with a particular external IP suddenly starts making repeated connections to it, I would investigate further.

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6. Example of a SOC Investigation

Suppose an endpoint suddenly starts sending a large amount of data to an unfamiliar external IP.

I would investigate:

1. Which device generated the traffic?
2. Which user was logged in?
3. What external IP was contacted?
4. Which port and protocol were used?
5. Which process created the connection?
6. Was there a DNS query for the destination domain?
7. Did the device show any other suspicious activity?

8. Was data being uploaded?

If the traffic is confirmed to be suspicious, it could indicate that a compromised system is communicating with an attacker or sending data outside the organization.

I would then check other logs and alerts to understand the full activity.

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7. Why Traffic Direction Matters

Inbound and outbound traffic can give different clues.

For example:

Inbound traffic can help identify:

- Scanning
- Brute-force attempts
- Unauthorized access attempts
- Exploitation attempts

Outbound traffic can help identify:

- Malware communication
- Data exfiltration
- Suspicious downloads
- Connections to malicious infrastructure
- Command and Control activity

However, I should not assume that all inbound traffic is dangerous or all outbound traffic is safe.

Normal systems constantly communicate in both directions.

The important thing is to look at the **source, destination, port, protocol, timing and context**.

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8. My Understanding

I understood that the main difference is simple:

Inbound = traffic coming into my system/network.

Outbound = traffic going out from my system/network.

As a SOC analyst, this distinction can help me understand what is happening during an investigation.

For example, suspicious inbound traffic could indicate someone trying to access a system, while suspicious outbound traffic could show that an already compromised system is communicating with an external server.

So, I should not look at the direction alone. I need to combine it with other information such as IP addresses, ports, protocols, processes and logs before deciding whether the traffic is suspicious.